ALGEBRA 2 · UNIT 8: SEQUENCES & SERIES

First Car Savings Plan

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan how to buy a first car. They model a growing savings plan with an arithmetic sequence and series, model car depreciation and account growth with geometric sequences and series, use sigma notation, analyze an infinite geometric series, and choose the best plan.

Students see arithmetic growth (steady raises in savings) next to geometric change (depreciation and interest). The series formulas answer real questions about totals.

Standards Addressed

- F-BF.2 Write arithmetic and geometric sequences
- A-SSE.4 Derive and use the geometric series formula
- F-IF.3 Sequences as functions
- F-LE.2 Construct sequences from descriptions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: how much does a used car cost? Part 1 arithmetic savings.
Day 2	Part 2 geometric models and series.
Day 3	Part 3 plan.

Materials

- Student handout
- Calculators or spreadsheets
- Optional: local used-car listings

How the Three Versions Differ

Level	What changes
CP2	Explicit rules and sums with formulas given, depreciation table, and a bouncing-ball series.
CP1	Series formulas, sigma notation, a monthly deposit account, and solving for n.
Honors	Derive the geometric series formula, derive the loan payment formula, and compare saving vs. borrowing.

Watch For (Common Misconceptions)

- Using n instead of n − 1 in the explicit formula.
- Using 0.15 instead of 0.85 for 15% depreciation.
- Thinking an infinite series always diverges.

Extension Ideas

- Use real loan rates.
- Build the plan in a spreadsheet.
- Research insurance costs for new drivers.

Answer Key – CP2

Part 1: Saving Up

#	Task	Answer
1	Write the first five terms and an explicit rule a_n.	50, 60, 70, 80, 90; $a_n = 50 + 10(n - 1)$
2	How much do you save in month 12?	$a_{12} = 50 + 110 = \$160$
3	Use $S_n = n(a_1 + a_n) \div 2$ to find the total saved in 24 months.	$a_{24} = 280$; $S_{24} = 24(50 + 280) \div 2 = \$3{,}960$

Part 2: Cars Lose Value

#	Task	Answer
4	Write a rule for the car's value after n years.	$V_n = 20{,}000(0.85)^n$
5	Find its value after 5 years.	$20{,}000(0.85)^5 \approx \$8{,}874.11$
6	A ball dropped from 6 ft bounces back to 60% of its height each time. List the first four heights.	6, 3.6, 2.16, 1.296 ft
7	The total distance the ball travels is $6 + 2(3.6 + 2.16 + \ldots)$. Use $S = a_1 \div (1 - r)$ for the sum in the parentheses and find the total distance.	$3.6 \div (1 - 0.6) = 9$; total $6 + 2(9) = 24$ ft

Part 3: Your Plan

#	Task	Answer
8	Find a used car's value (or use a 5-year-old $20,000 car). Can your 24-month savings buy it?	$3,960 is not enough for $8,874 (answers vary for other cars).
9	Change your plan (start amount or monthly increase) so you can afford the car in 24 months.	Answers vary.
10	Write your plan.	Answers vary.

Answer Key – CP1

Part 1: Saving Up

#	Task	Answer
1	Write the total for 24 months in sigma notation and evaluate it.	Σ (n = 1 to 24) of [50 + 10(n − 1)] = $3,960
2	After how many months will the total first reach $5,000?	$n \div 2 \cdot (100 + 10(n-1)) \geq 5000 \rightarrow n^2 + 9n - 1000 \geq 0 \rightarrow n \geq 27.4$: month 28 ($S_{27}$ = 4,860, S_{28} = 5,180)
3	Plan B: deposit $200 each month into an account that pays 0.5% per month. After 24 months, the value is $200(1.005^{24} - 1) \div 0.005$. Evaluate it and…	≈ $5,086.39. Each deposit grows by a factor of 1.005 for a different number of months, so the values form a geometric series.

Part 2: Cars & Bounces

#	Task	Answer
4	The car loses 15% per year from $20,000. In which year is it first worth less than $10,000?	$0.85^n < 0.5 \rightarrow n > \log 0.5 \div \log 0.85 \approx 4.27$: year 5
5	Write the depreciation as a recursive rule.	$V_0 = 20{,}000$; $V_n = 0.85V_{n-1}$
6	Find the total distance traveled by a ball dropped from 6 ft that rebounds to 60% each bounce.	24 ft

Part 3: Your Plan

#	Task	Answer
7	Compare the two plans after 24 months.	Plan B ($5,086) > Plan A ($3,960)
8	Choose a car and year and decide which plan (or combination) works.	Answers vary.
9	Write the plan with sigma notation for one total.	Answers vary.

Answer Key – Honors

Part 1: Derivations

#	Task	Answer
1	Derive $S_n = a(1 - r^n) \div (1 - r)$ by computing $S - rS$.	$S - rS = a - ar^n \to S(1 - r) = a(1 - r^n) \to S_n = a(1 - r^n) \div (1 - r)$
2	Use it to show that 24 monthly deposits of \$200 at 0.5% per month grow to $200(1.005^{24} - 1) \div 0.005$. Evaluate.	$a = 200$, $r = 1.005$, $n = 24$: $200(1.005^{24} - 1) \div 0.005 \approx$ \$5,086.39
3	Explain why the sum of an infinite geometric series exists only when $\|r\| < 1$.	$r^n \to 0$ only if $\|r\| < 1$; otherwise the terms don't shrink and the sum grows without bound (or oscillates).

Part 2: Borrowing

#	Task	Answer
4	The present values of the payments form a geometric series: $L = M(1 + r)^{-1} + M(1 + r)^{-2} + \ldots + M(1 + r)^{-n}$. Show that $M = rL \div (1 - (1 + r)^{-n})$.	Sum with first term $M(1 + r)^{-1}$ and ratio $(1 + r)^{-1}$: $L = M(1 - (1 + r)^{-n}) \div r$, so $M = rL \div (1 - (1 + r)^{-n})$.
5	Find the monthly payment for \$15,000 at 0.5% per month for 48 months, and the total interest paid.	$M = 75 \div (1 - 1.005^{-48}) \approx$ \$352.28; total paid ≈ \$16,909.22; interest ≈ \$1,909.22
6	Compare: save \$352.28 per month for 48 months at 0.5% per month first, then buy. How much would you have?	$352.28(1.005^{48} - 1) \div 0.005 \approx$ \$19,057

Part 3: Decision

#	Task	Answer
7	Compare saving first with borrowing now, using your numbers.	Answers vary.
8	Consider depreciation: the car loses 15% per year. Include it in your decision.	Answers vary.
9	Write the final plan.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Models are correct sequences or series, and the decision uses the totals correctly. Goes beyond the minimum with a creative, realistic choice.	Models are correct sequences or series, and the decision uses the totals correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes savings and depreciation models, totals using series, and a final decision with a timeline. Polished, organized, and visually engaging.	Plan includes savings and depreciation models, totals using series, and a final decision with a timeline. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.